

The vector equation of a line

Cambridge insert: a point and a direction give every point of a line — and intersection, parallelism and skewness all become arithmetic.

LESSON 14–16

3 × 40 MIN

GEOMETRY 11, §8 (EXTENSION)

P3 · 9.1–9.3

LESSON CLOCK

14'

22'

24'

26'

30'

19'

A point and a direction	14 min
From two points	22 min
Do two lines meet?	24 min
Parallel, intersecting or skew	26 min
The angle	30 min
Practice and homework	19 min

LEARNING OBJECTIVES

- Write the vector equation of a line from a point and a direction.
- Write it from two points, and convert to parametric form.
- Decide whether two lines meet, are parallel, or are skew.
- Find the angle between two lines and the point of intersection.

TEXTBOOK

National	pp. 77–84
Cambridge	Pure Mathematics 2 & 3, pp. 190–208
Workbook	P3 Exercise 9A–9C

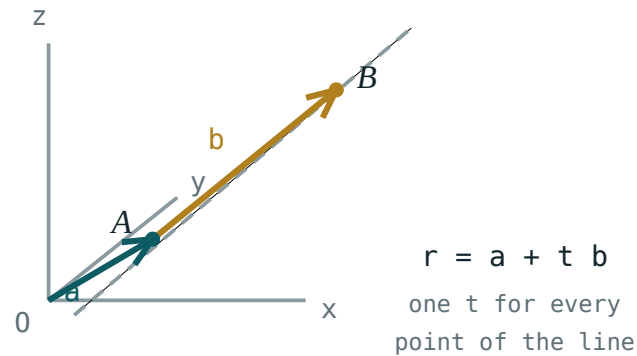
01

Explanation

A point and a direction

Start at a known point A with position vector a , and travel any multiple of a direction d :

$$r = a + t d, t \in \mathbb{R}$$



A fixed point, a direction arrow, and the whole line traced out as t runs over the reals.

Each value of t gives one point; $t = 0$ gives A itself. In components, with $a = (1, 2, 3)$ and $d = (2, -1, 4)$:

$$x = 1 + 2t, y = 2 - t, z = 3 + 4t$$

THE EQUATION IS NOT UNIQUE

Any point of the line may play a , and any non-zero multiple of d may play the direction. $r = (3, 1, 7) + s(4, -2, 8)$ is the **same line**. Never conclude two lines differ just because the equations look different.

From two points

Take one point as a and the vector between them as d :

$$r = a + t(b - a)$$

Through $A(1, 0, 2)$ and $B(4, 6, 8)$: $d = (3, 6, 6)$, or more simply $(1, 2, 2)$, so $r = (1, 0, 2) + t(1, 2, 2)$.

SIMPLIFY THE DIRECTION

Dividing d by a common factor changes nothing but makes every later calculation easier — and makes parallel lines obvious at a glance.

Do two lines meet?

THE PROCEDURE

Set the two position vectors equal, using **different** parameters t and s . That gives three equations in two unknowns. Solve two of them; substitute into the third. Consistent $\Rightarrow$ they meet. Inconsistent $\Rightarrow$ they do not.

$$\ell_1 : r = (1,0,2) + t(1,2,2) \text{ and } \ell_2 : r = (2,5,3) + s(0,1,-1):$$

$$1 + t = 2 \Rightarrow t = 1; 2t = 5 + s \Rightarrow s = -3; \text{ check: } 2 + 2t = 4, 3 - s = 6 \quad \times$$

The third equation fails, so the lines do not meet. Their directions are not parallel either, so they are **skew**.

USE TWO DIFFERENT LETTERS

Writing t in both equations forces the two lines to reach the meeting point “at the same time”, which is an extra condition the geometry never imposed. It is the single most common error in this topic.

Classifying, and the angle

DIRECTIONS	SYSTEM	CONCLUSION
parallel	—	parallel (or identical, if a point is shared)
not parallel	consistent	they intersect — one point
not parallel	inconsistent	skew

The **angle** between two lines is the angle between their directions, from the scalar product — and it is defined even for skew lines, where the lines themselves never meet:

$$\cos \theta = \frac{|\mathbf{d}_1 \cdot \mathbf{d}_2|}{|\mathbf{d}_1||\mathbf{d}_2|}$$

The modulus in the numerator forces the acute angle, which is the conventional answer.

02 Worked examples

EXAMPLE 1 Find the vector equation of the line through $A(2, -1, 3)$ and $B(4, 3, 7)$.

① $d = b - a = (2, 4, 4)$

Simplify to $(1, 2, 2)$.

② $r = (2, -1, 3) + t(1, 2, 2)$

③ Parametric: $x = 2 + t, y = -1 + 2t, z = 3 + 2t$.

ANSWER

$$r = (2, -1, 3) + t(1, 2, 2)$$

EXAMPLE 2 Do $\ell_1 : r = (1, 1, 1) + t(2, 1, 3)$ and $\ell_2 : r = (3, 2, 4) + s(1, -1, 2)$ meet?

① $1 + 2t = 3 + s$ and $1 + t = 2 - s$

② Adding: $2 + 3t = 5 \Rightarrow t = 1$, then $s = 0$.

③ Third: $1 + 3t = 4$ and $4 + 2s = 4$.

Both give 4 ✓

④ Point: $(3, 2, 4)$.

ANSWER

Yes, at $(3, 2, 4)$

EXAMPLE 3 Find the acute angle between the lines with directions $(1, 2, 2)$ and $(2, -1, 2)$.

$$① \quad d_1 \cdot d_2 = 2 - 2 + 4 = 4$$

$$② \quad |d_1| = |d_2| = 3$$

$$③ \quad \cos \theta = \frac{4}{9}$$

$$④ \quad \theta \approx 63.6^\circ$$

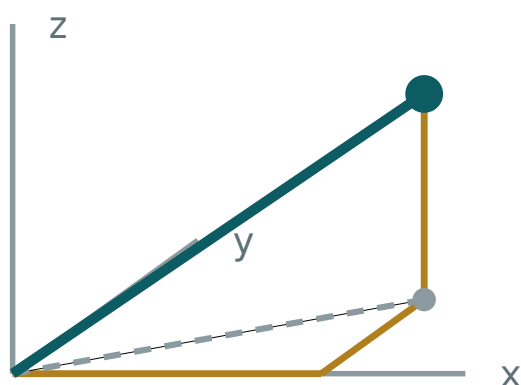
ANSWER

$$\approx 63.6^\circ$$

03 Interactive model

Vary t and watch the point travel along the line; then add a second line and look for a crossing.

A line in space



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Vector equation of a line	To'g'ri chiziq vektor tenglamasi	Векторное уравнение прямой
Direction vector	Yo'naltiruvchi vektor	Направляющий вектор
Parameter	Parametr	Параметр
Parametric form	Parametrik shakl	Параметрический вид
Point of intersection	Kesishish nuqtasi	Точка пересечения
Skew lines	Ayqash chiziqlar	Скрещивающиеся прямые
Position vector	Radius-vektor	Радиус-вектор
Consistent system	Birgalikdagi sistema	Совместная система
Angle between lines	Chiziqlar orasidagi burchak	Угол между прямыми

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

In $r = a + td$, d is:

a point

the direction

the length

a parameter

The line through A and B is:

$r = a + tb$

$r = a + t(b - a)$

$r = ab$

$r = a - b$

When two lines are compared you must use:

the same parameter

two different parameters

no parameter

three parameters

Non-parallel directions with an inconsistent system means:

intersecting

parallel

skew

identical

The angle between lines uses:

$|d_1 \cdot d_2|$

$d_1 \cdot d_2$

$|d_1| + |d_2|$

the points

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Line through $(0,0,0)$ with direction $(1,1,1)$

ANSWER $r = t(1,1,1)$

2. Line through $(1,2,3)$ with direction $(0,0,1)$

ANSWER $r = (1,2,3) + t(0,0,1)$

3. Is $(2,4,6)$ on $r = t(1,2,3)$?

ANSWER yes, $t = 2$

4. Is $(1,1,2)$ on $r = t(1,1,1)$?

ANSWER no

5. Direction of the line through $(1,0,0)$ and $(3,2,4)$

ANSWER $(2,2,4)$ or $(1,1,2)$

6. Are $(1,2,2)$ and $(2,4,4)$ parallel directions?

ANSWER yes

7. Value of t giving the point a itself

ANSWER $t = 0$

Medium · the standard question

7 PROBLEMS

1. Line through $A(2, -1, 3)$ and $B(4, 3, 7)$

ANSWER $r = (2, -1, 3) + t(1, 2, 2)$

2. Parametric form of that line

ANSWER $x = 2 + t, y = -1 + 2t, z = 3 + 2t$

3. Do $r = (1, 1, 1) + t(2, 1, 3)$ and $r = (3, 2, 4) + s(1, -1, 2)$ meet?

ANSWER yes, at $(3, 2, 4)$

4. Angle between directions $(1, 2, 2)$ and $(2, -1, 2)$

ANSWER $\approx 63.6^\circ$

5. Angle between $(1, 0, 0)$ and $(1, 1, 0)$

ANSWER 45°

6. Are $r = (0, 0, 0) + t(1, 2, 3)$ and $r = (1, 2, 3) + s(2, 4, 6)$ parallel?

ANSWER yes — the same line

7. Where does $r = (1, 2, 3) + t(1, 1, 1)$ meet the plane $z = 0$?

ANSWER $(-2, -1, 0)$

Hard · stretch and reason

7 PROBLEMS

1. Show $r = (1,0,2) + t(1,2,2)$ and $r = (2,5,3) + s(0,1,-1)$ are skew

ANSWER Directions not parallel; the system is inconsistent

2. Find where $r = (2,0,1) + t(1,1,2)$ meets $r = (4,2,5) + s(2,1,1)$

ANSWER $(4,2,5)$

3. Find the angle between the lines in the previous question

ANSWER $\approx 33.6^\circ$

4. Find λ so $(3, \lambda, 6)$ lies on $r = (1,0,2) + t(1,2,2)$

ANSWER $\lambda = 4$

5. Find the point of $r = t(1,2,2)$ closest to $(3,0,0)$

ANSWER $(\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$

6. Show the lines through $(0,0,0), (1,1,1)$ and $(1,0,0), (0,1,1)$ intersect

ANSWER At $(0.5, 0.5, 0.5)$

7. Write the equation of the x -axis in vector form

ANSWER $r = t(1,0,0)$

07 Homework

Homework — 6 tasks

Use different letters for the two parameters, every time.

- Find the vector equation of the line through $A(1, -2, 4)$ and $B(5, 2, 12)$.
- Write that line in parametric form and find the point with $t = 3$.
- Decide whether $r = (1,2,0) + t(1,1,1)$ and $r = (3,4,2) + s(2,1,3)$ meet, and if so where.
- Find the acute angle between the directions $(2, 1, -2)$ and $(1, 2, 2)$.
- Show that $r = (0,0,0) + t(1,1,0)$ and $r = (0,1,1) + s(1,0,0)$ are skew.

6. Find where $r = (2, 1, 5) + t(1, -1, 2)$ crosses the plane $y = 0$.